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Development of an LLM Multi-Agent Simulation and Learning Support Platform for Designing Teacher-in-the-Loop Differentiated Instruction

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01. Introduction

- **Need for study** : Although elementary school classrooms aim to consider student diversity, there are difficulties in environments where there is only one teacher and many students.
- **Advancement of LLMs** : Generative AI endowed with personas via system prompts can now sophisticatedly simulate diverse student responses.
- **Evolution of Edutech** : learning-supported Edtech offers the possibility of personalized education by reorganizing learning materials and problems

RESEARCH GOAL
Developing a platform that prepares lessons through pre-simulation using teacher-student multi-agents, and provides and manages individualized learning materials.

02. Theoretical Background

- **Generative Agents Simluation** : Beyond a single agent, it enable interactions between teachers and students and between students, vividly recreating the dynamics of the real classroom.
- **Teacher-in-the-Loop AI** : Teachers retain final decision and editing authority over AI-generated responses and items, ensuring educational reliability and foregrounding instructional design expertise.
- **Differentiated Instruction** :
  - learning effectiveness by adjusting instuction and assessments considering students' preparation, interest, and learning styles.
  - This is based on Zone of Proximal Development (ZPD) theory and supports optimal learning pace for each student through an AI-based automated assessment system.

03. System Architecture & Features

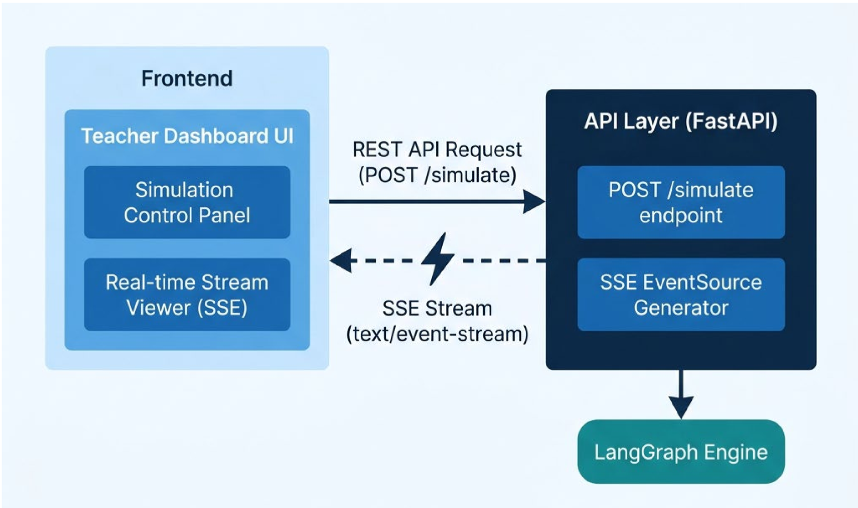

Fig1. Frontend-API Communication via REST and SSE

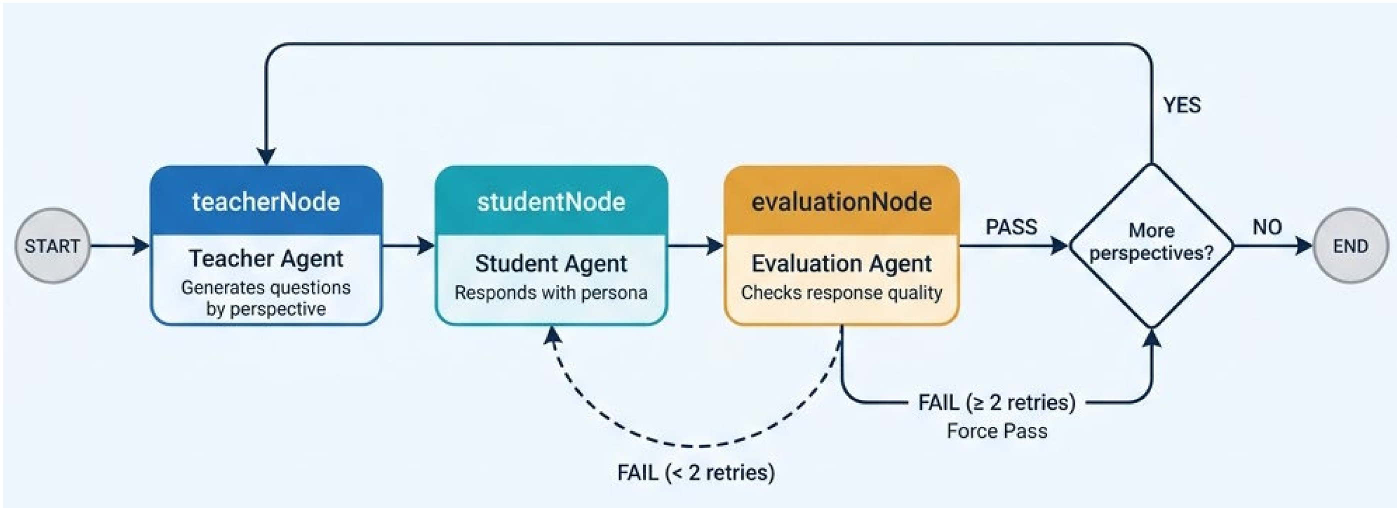

Fig2. Multi-Agent Simulation Pipeline via LangGraph

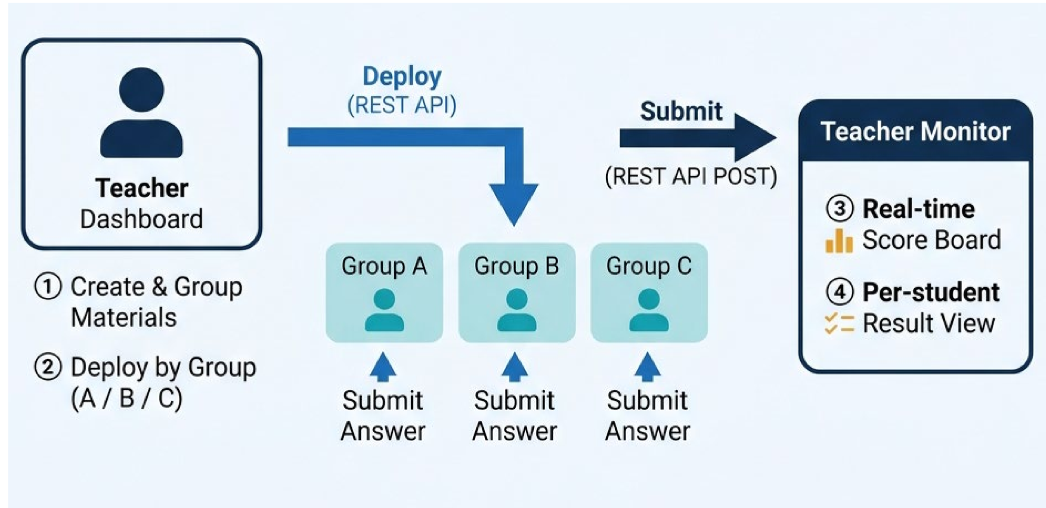

Fig3. Group Deployment and Real-Time Monitoring Flow

Subject-specific Simluation Dimensions

- Science :
  Misconceptions | Real-life connections | Lab Safety | Learning connections
- English :
  Phonics | Advanced dialogue & expression | Real life application | Grammar

Multi-Agent Architecture

Teacher Agent
Analyzes achievement standards, topics, and learning activities textbook data to generate multidimensional questions.

Student Agent
Generates multidimensional questions according to its assigned perspective (persona).

Evalution Agent
Rule-based check of structural quality; on failure, requests regeneration (forced pass after 2 retries).

Teacher Dashboard

- Run simulations and generate lesson cue cards & lesson materials.
- Create materials (quiz / media) from a repository and combine by proficiency level.
- Distribute materials for groups or individual students.
- Submission/score status grids (by class/subject), individual analysis.

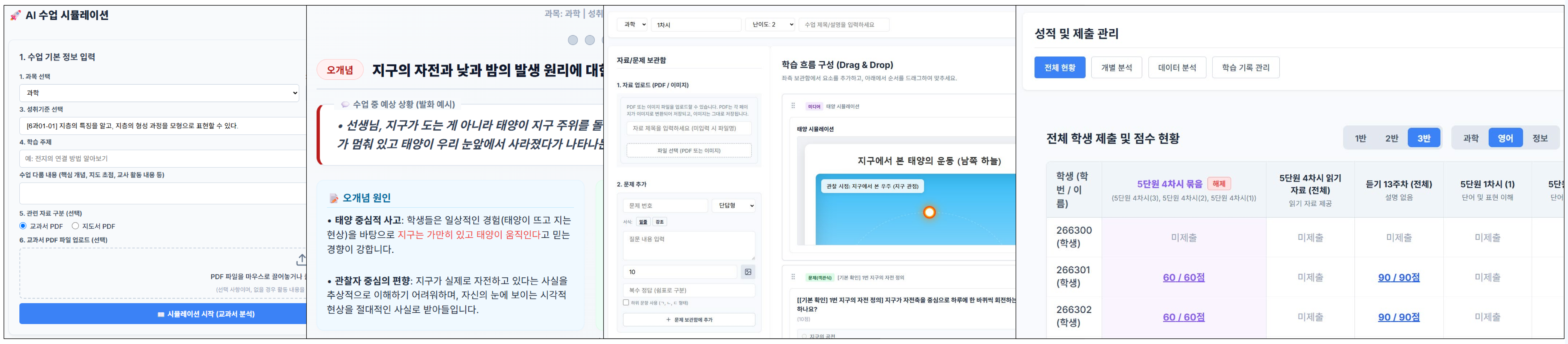

Fig4. Teacher dashboard screen simulation creation, created CUE card, create and combine learning materials, submission and score status

Student Workspace

- Receive media(link, image, HTML simulation) materials and quiz
- Solve quiz and automated-assessment.
  (When resolving the quiz, the order of the quiz gets mixed up.)

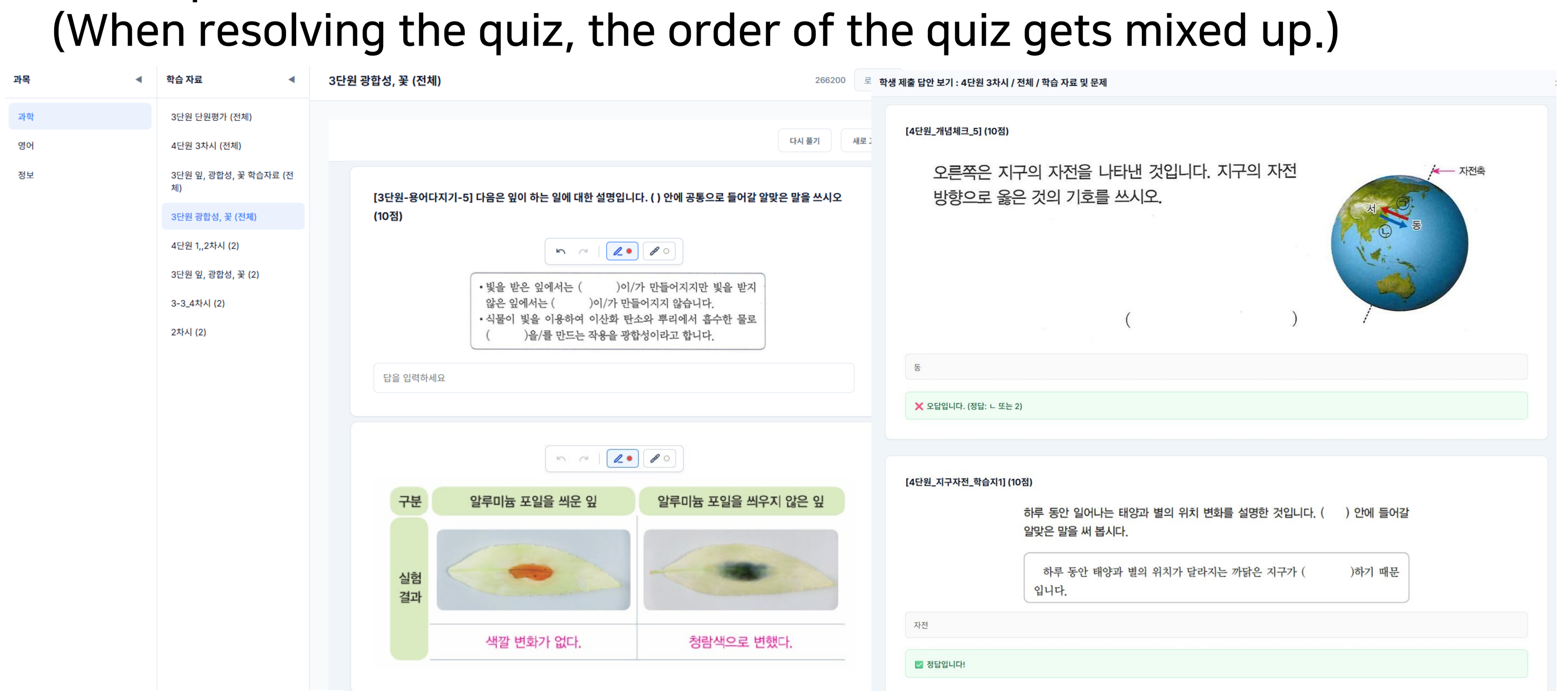

Fig5. student workspace and automated-assesment

Lesson Flow

- Lesson Preparation
  - Analyze achievement standards → LLM simulation (generate cue cards) → create material & quiz → combine differentiated material & quiz
- Lesson Execution
  - Teach with differentiated materials → solve the quiz (automated assessment) → real-time monitoring and individual feedback.
- Post lesson
  - Accumulate of individualized learning data → integrate results into the next lesson.

04. Technologies

LLM	Gemini-3.1-Flash-lite
Agent Framework	LangChain + LangGraph
Frontend	Next.js(React)
Database	SQLite(Prisma ORM)
Real-Time	Server-Sent Events (SSE)

05. Conclusion & Future work

- **Significance**
  : By using LLMs in lesson preparation to generate materials that account for diverse student responses, and by pairing differentiated materials with automated assessment, the system reduces teachers' in-class cognitive load and demonstrates the feasibility of differentiated instruction in elementary classrooms.
- **Future Directions**
  : Move beyond the current rule-based distribution by integrating accumulated student item-response data into an LLM-based customized item-recommendation algorithm.

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